AeroSentinel: Autonomous CubeSat Fault Recovery

An Unbreakable RL Environment with Deterministic Orbital Multi-Physics

- Presenter: Devesh Srivastava (Avionics Head, IIT BHU Rocketry Division)
- Domain: Autonomous Satellite Fault Diagnosis & Flight Software Remediation
- Challenge: Build an RL environment that cannot be broken, cheated, or bypassed
- Key Innovation: Authoritative 95-minute orbital physics engine with unified episode state

The Problem: Why Spacecraft Autonomy Matters

High Latency, Ground Pass Windows, and Catastrophic Time Bounds

- LEO spacecraft operate with only 8 to 12 minutes of ground station contact per 95-minute orbit
- Critical failures (battery eclipse collapse, thermal runaway, ADCS tumble) develop in < 15 minutes
- Ground-in-the-loop intervention is impossible during deep space and eclipse passes
- Autonomous onboard fault recovery is required to diagnose telemetry and restore nominal state

What Weak Environments Look Like vs. AeroSentinel

Moving Beyond Synthetic Puzzles to Real Physical Invariants

- Weak Environments: Static 'fix-the-bug' scripts with visible unit tests and pass/fail equality
- Weak Verifiers: LLM-as-judge or substring checks easily gamed by mock comments and sycophancy
- AeroSentinel: Real coupled physics — power, thermal, attitude, and RF link closure interact
- Zero Answer Leakage: Prompts provide flight manuals and telemetry logs, never solution numbers

The 95-Minute LEO Orbital Physics Engine

Coupled Multi-Subsystem State Transitions

- Electrical Power (EPS): Solar generation (0.06 m2 GaAs, 28% eff), 40 Wh Li-ion, 3.10V cutoff
- Attitude Control (ADCS): 3-axis reaction wheel momentum bias, precession rate, sun pointing
- Thermal Control (TCS): Radiative cooling to 4K space, internal dissipation, thermostat hysteresis
- RF Communications (COMMS): UHF 437.5 MHz Friis link budget, 141.5 dB FSPL, baud-rate bounded throughput

Unbreakable Security & Isolation Architecture

Zero Trust Sandbox with Out-of-Band Evaluation

- Isolated Verifier: Verifier and orbital simulator run in Harness container only (zero sandbox access)
- Safe Script Evaluation: Untrusted agent Python code evaluated via unprivileged sandbox worker (UID 1001)
- Dynamic Token Auth: Privileged endpoints (/reset, /state, /fetch_configs, /eval_script) require 256-bit token
- Monotonic Time Limits: Hard 600-second execution deadline across model calls, 429 retries, and tools
- Robust Limits: 20s tool timeouts, 4000-char clipping, 48K context window trimming

Five Realistic Multi-Physics Flight Tasks

Graduated Difficulty Across Two Tiers with Seeded Variations

- Task 1 (Easy): EPS Eclipse Recovery — solar calibration & undervoltage protection
- Task 2 (Easy): ADCS Detumble & Sun-Pointing — reaction wheel momentum bias elimination
- Task 3 (Medium): Battery Thermal Preservation — thermostat setpoint & runaway prevention
- Task 4 (Medium): RF Downlink Closure — transmitter EIRP, antenna gain, dynamic link budget
- Task 5 (Medium/Hard Seeded): Safe Mode Cascade — coupled power-thermal-ADCS emergency recovery

Programmatic Outcome-Based Scoring Model

Rewarding Mission Survival and Safe Operations

- Reward Formula: Power_Balance (0.35) + Thermal_Safety (0.25) + ADCS (0.20) + Comms (0.20)
- Catastrophic Zeroing: Irreversible violations (bus collapse <= 3.10V, temp > 50°C) permanently zero score
- Unified Trajectory: Scoring evaluates actual operational trajectory plus completed orbit
- Objective & Deterministic: Programmatic telemetry evaluation — zero LLM-as-judge variance

Interactive Orbital Telemetry Progression

Observing Dynamic Consequences Across Simulated Intervals

- Interactive Tool: step_orbit allows agent to advance orbit and observe intermediate consequences
- Live Telemetry: Generates real-time telemetry logs at telemetry/live_telemetry.csv
- Demonstrated: Live model gemini-3.5-flash-lite actively called step_orbit to verify fixes
- Defenses: Subprocess environment scrubbed; directory traversal blocked via safe_path()

Live Run Performance & Baseline Gates

Exceeding All Competition Thresholds

- Oracle Baseline: 1.00 / 1.00 across all 5 tasks (Requirement: >= 0.95)
- No-Op Baseline: 0.00 / 1.00 across all 5 tasks (Requirement: <= 0.05)
- Gate Execution Time: 0.05 seconds for all 5 tasks combined (Budget: 120 seconds)
- Real Model Evidence: Genuine trajectories recorded for gemini-3.5-flash-lite achieving 1.00
- Ready for Run A (GPT-Luna) and Run B (Undisclosed Model Stress Test)

Conclusion: Flight-Ready Space Software

Bridging Autonomous AI Agents and Aerospace Engineering

- AeroSentinel demonstrates that RL environments can have physical depth, fidelity, and ironclad security
- Grounded in real rocketry avionics software practices from IIT BHU
- GitHub Repository: github.com/devesh3001/Aerosentinel (100% complete, tested, and pushed)
- Submission Packages: Astro.zip and Aerosentinel.zip ready for evaluation